

Prompting Fundamentals on Real Tasks (Code: FL-02)

Track: General AI Fluency
Type: Assignment Deliverable

Phase: Foundations (Week 2)
Workload: 6 Hours

Task Selected from FL-01 Audit: Synthesizing Massive Technical Documentation / Engineering Status Reports into Actionable Executive Briefings.
Core Challenge: Technical teams produce granular status updates full of query metrics and infrastructure noise. Non-technical leadership needs a concise snapshot focusing on Project Health, Active Blockers, and Milestone Timelines.

VERSION 0: NAIVE BASELINE (NO TECHNIQUES APPLIED)

Prompt: "Summarize this weekly engineering report for executives: [Paste Report]"

RAW OUTPUT:
"The team completed 14 database query refactors and upgraded PostgreSQL to v15.2. API latency dropped by 120ms. The backend team spent 25 hours on CSS fixes and auth middleware. Meanwhile, AWS deployment was paused due to cost overruns."

Output Note & Analysis: Dumps raw technical metrics without explaining why the 120ms latency drop matters or highlighting that the AWS pause is a major financial blocker. Fails to provide executive-level synthesis.

VERSION 1: ROLE ASSIGNMENT (PERSONA PIVOT)

Prompt: "You are an experienced Chief Technology Officer (CTO). Summarize this weekly engineering report for non-technical C-suite executives: [Paste Report]"

RAW OUTPUT:
"Engineering progress remains steady this week. We made notable achievements in core infrastructure performance, resulting in a ~15% boost in platform response speed. However, cloud infrastructure migration is temporarily held pending cost optimizations."

Output Note & Analysis: Tone improved immediately from passive logs to executive voice, translating query refactors into "15% platform speed boost." Still structured as an unstructured paragraph rather than a scannable dashboard.

VERSION 2: CONTEXT AND MOTIVATION (HIGH-STAKES CONSTRAINTS)

Prompt: "You are an experienced CTO. Summarize this engineering report for executives who have exactly 30 seconds before a board meeting. Focus strictly on strategic blockers, resource needs, and key launch dates: [Paste Report]"

RAW OUTPUT:
"• Critical Blocker: AWS Cloud Migration paused due to budget overruns. Requires CFO sign-off.
• Key Win: Platform responsiveness improved by 15%.
• Launch Impact: Target launch date holds at Sept 15, conditional on budget approval."

Output Note & Analysis: The 30-second pre-meeting context forced the model to strip out minor maintenance details (CSS fixes) and pull critical financial blockers straight to the top.

VERSION 3: FEW-SHOT EXAMPLES (PATTERN ANCHORING)

Prompt: "You are an experienced CTO. Summarize this engineering report for executives.
Example Input: 'Fixed 12 backend bugs and delayed migration due to AWS budget limits.'
Example Output:
- Highlights: Core backend stability improved.
- Blockers: Infrastructure migration paused (Budget Approval Required).
Task Input: [Paste Report]"

RAW OUTPUT:

"- Highlights: Infrastructure speed improved by 15%.
- Blockers: AWS migration halted due to cloud spend cap limits.
- Launch Status: Target release holding firm for Sept 15."

Output Note & Analysis: Providing a concrete exemplar locked the model into consistent bullet lengths and removed introductory conversational padding completely.

VERSION 4: OUTPUT STRUCTURE (SCHEMA ENFORCEMENT)

Prompt: "You are an experienced CTO summarizing an engineering report for executives. Format your response strictly using this Markdown layout:
Project Health Status
Top Business Impact Wins
Critical Blockers & Action Items
Launch Timelines
Report: [Paste Report]"

RAW OUTPUT:

"## Project Health Status: Yellow (Deployment paused pending budget review)
Top Business Impact Wins: Reduced application latency by 15%, improving checkout flow speed.
Critical Blockers & Action Items: AWS deployment stalled; requires executive budget reallocation.
Launch Timelines: Target Date: September 15 (On track)."

Output Note & Analysis: Layout consistency achieved. The fixed schema guarantees predictable navigation for leadership review week over week.

VERSION 5: STEP DECOMPOSITION (CHAIN-OF-THOUGHT REASONING)

Prompt: "You are an experienced CTO summarizing an engineering report for executives. Work through these steps before generating output:
Step 1: Translate technical updates into clear business value metrics.
Step 2: Flag active budget, timeline, or resource risks requiring leadership intervention.
Step 3: Determine overall project health status (Green = On track, Yellow = Risks present, Red = Launch delayed).
Step 4: Format output using the approved structure.
[Structure + Report attached]"

RAW OUTPUT:

"## Project Health Status: YELLOW (Core build is stable, but cloud budget friction requires immediate CFO intervention)
Top Business Impact Wins: System responsiveness increased by 15% (directly impacts customer retention).
Critical Blockers & Action Items: AWS cloud spend threshold exceeded by 12%. Needs CFO override to resume migration.
Launch Timelines: Target Date: Sept 15 (Risk level: Low if budget cleared by end-of-week)."

Output Note & Analysis: Step decomposition eliminated arbitrary ratings. The 'YELLOW' status was logically deduced from the financial blocker isolated in Step 2, making the summary airtight.

3. CROSS-MODEL EVALUATION (CLAUDE 3.5 SONNET VS. CHATGPT GPT-4O)

Both frontier models were executed on Version 5 using identical engineering reports:

Evaluation Dimension	Claude 3.5 Sonnet	ChatGPT (GPT-4o)
Tone & Executive Voice	Deeply authoritative, measured, and natural. Reads like an experienced in-house CTO communicating with peers.	Fast, punchy, and structured, but leans slightly formal or template-driven in phrasing.
Technical Translation	High fidelity. Preserved critical infrastructure context while explaining direct business implications.	Solid business abstraction, though occasionally omitted technical dependencies (e.g., specific AWS constraints).
Structural Adherence	Followed Markdown headers and step constraints with zero deviation or extra conversational preamble.	Followed headings accurately, with a minor tendency to over-bold keywords inside bullet points.
Specific Failure Points	Can write slightly dense rationale sentences unless strict character limits are provided.	Tends to assume optimistic timelines for blockers unless explicitly forced to mark risk status.

4. FINAL REUSABLE PROMPT TEMPLATE (CONTEXT-FREE)

Production-ready template structured for any team or domain without personal or project dependencies:

[ROLE & OBJECTIVE]

You are a senior [INSERT ROLE, e.g., CTO / Engineering Lead / Product VP]. Summarize the attached status report for [INSERT AUDIENCE, e.g., Executive Leadership / Board Members].

[CONTEXT & CONSTRAINTS]

The audience needs a 30-second briefing to evaluate operational health, resource bottlenecks, and key deliverables. Avoid technical jargon without business context.

[EXECUTION WORKFLOW]

1. Scan the report and translate technical activities into clear business and operational impact metrics.
2. Isolate active blockers, budget risks, or resource bottlenecks requiring executive intervention.
3. Assign an overall Health Rating (Green / Yellow / Red) with a 1-sentence justification.
4. Output using ONLY the structured schema below.

[REQUIRED OUTPUT SCHEMA]

```
## Overall Status: [Green / Yellow / Red] - [1-Sentence Rationale]
## Major Accomplishments & Business Impact
* [Win 1 + Operational Value]
* [Win 2 + Operational Value]
## Strategic Blockers & Actions Required
* [Blocker 1] - Action Needed: [Specific Request / Owner]
## Key Milestones & Target Dates
* [Date]: [Deliverable Status]
```

[SOURCE DATA]

[PASTE REPORT / LOGS HERE]